Assignment 2: Newton–Raphson Load Flow Programming Exercise

IIT Delhi – Abu Dhabi

Course Code: AENL226

Course Name: Power Electronics and Power Systems

Unless otherwise stated, all quantities are in per unit on a 100 MVA base.

Use the following conventions: the listed b is the total line-charging susceptance, so use $jb/2$ at each end of a line; all transformer tap ratios are 1.0; use a flat start; and terminate when the maximum power mismatch is below 10^{-8} p.u.

Question 1: 4-Bus System

The system consists of one slack bus, one PV bus, and two PQ buses. Use the following bus and branch data.

Bus data

Bus	Type	P_D MW	Q_D MVAr	P_G MW	$\|V\|$ p.u.	$Q_{G,\min}$ MVAr	$Q_{G,\max}$ MVAr
1	Slack	0	0	unknown	1.040	–	–
2	PV	20	10	70	1.010	-30	40
3	PQ	40	15	0	–	–	–
4	PQ	30	10	0	–	–	–

Take $\delta_1 = 0°$. For PQ buses, use $|V| = 1.0$ p.u. only as the initial value.

Branch data

No.	From	To	r (p.u.)	x (p.u.)	b (p.u.)
1	1	2	0.0200	0.0600	0.0300
2	2	3	0.0250	0.0750	0.0400
3	3	4	0.0300	0.0900	0.0300

Question. Solve using Newton–Raphson load flow method. Enforce PV-bus reactive-power limits with PV-to-PQ conversion if required. Report the converged bus voltages, generator powers, branch power flows, total real-power loss, iteration count, and final mismatch.

Question 2: 30-Bus System

The system consists of one slack bus, five PV buses, and twenty-four PQ buses. Use the following bus data.

Bus data

Bus	Type	P_D MW	Q_D MVAr	P_G MW	$\|V\|$ p.u.	$Q_{G,\min}$ MVAr	$Q_{G,\max}$ MVAr
1	Slack	0.0	0.0	unknown	1.050	–	–
2	PV	17.4	10.2	32.0	1.040	-40	80
3	PQ	1.9	1.0	0.0	–	–	–
4	PQ	6.1	1.3	0.0	–	–	–
5	PV	75.4	15.2	24.0	1.010	-40	60
6	PQ	0.0	0.0	0.0	–	–	–
7	PQ	18.2	8.7	0.0	–	–	–
8	PV	24.0	24.0	28.0	1.010	-40	60
9	PQ	0.0	0.0	0.0	–	–	–
10	PQ	4.6	1.6	0.0	–	–	–
11	PV	0.0	0.0	16.0	1.020	-30	50
12	PQ	9.0	6.0	0.0	–	–	–
13	PV	0.0	0.0	16.0	1.010	-30	50
14	PQ	5.0	1.3	0.0	–	–	–
15	PQ	6.6	2.0	0.0	–	–	–
16	PQ	2.8	1.4	0.0	–	–	–
17	PQ	7.2	4.6	0.0	–	–	–
18	PQ	2.6	0.7	0.0	–	–	–
19	PQ	7.6	2.7	0.0	–	–	–
20	PQ	1.8	0.6	0.0	–	–	–
21	PQ	14.0	9.0	0.0	–	–	–
22	PQ	0.0	0.0	0.0	–	–	–
23	PQ	2.6	1.3	0.0	–	–	–
24	PQ	7.0	5.4	0.0	–	–	–
25	PQ	0.0	0.0	0.0	–	–	–
26	PQ	2.8	1.8	0.0	–	–	–
27	PQ	0.0	0.0	0.0	–	–	–
28	PQ	0.0	0.0	0.0	–	–	–
29	PQ	1.9	0.7	0.0	–	–	–
30	PQ	8.5	1.5	0.0	–	–	–

Take $\delta_1 = 0°$. For all PQ buses, use $|V| = 1.0$ p.u. only as the initial value.

Branch data

No.	From	To	r (p.u.)	x (p.u.)	b (p.u.)
1	1	2	0.0200	0.0600	0.0300
2	1	3	0.0500	0.1900	0.0200
3	2	4	0.0600	0.1700	0.0200
4	3	4	0.0100	0.0400	0.0000
5	2	5	0.0500	0.2000	0.0200
6	2	6	0.0600	0.1800	0.0200
7	4	6	0.0100	0.0400	0.0000
8	5	7	0.0500	0.1200	0.0100
9	6	7	0.0300	0.0800	0.0100
10	6	8	0.0100	0.0400	0.0000
11	6	9	0.0000	0.2100	0.0000
12	6	10	0.0000	0.5600	0.0000
13	9	11	0.0000	0.2100	0.0000
14	9	10	0.0000	0.1100	0.0000
15	4	12	0.0000	0.2600	0.0000
16	12	13	0.0000	0.1400	0.0000

No.	From	To	r (p.u.)	x (p.u.)	b (p.u.)
17	12	14	0.1200	0.2600	0.0000
18	12	15	0.0700	0.1300	0.0000
19	12	16	0.0900	0.2000	0.0000
20	14	15	0.2200	0.2000	0.0000
21	16	17	0.0800	0.1900	0.0000
22	15	18	0.1100	0.2200	0.0000
23	18	19	0.0600	0.1300	0.0000
24	19	20	0.0300	0.0700	0.0000
25	10	20	0.0900	0.2100	0.0000
26	10	17	0.0300	0.0800	0.0000
27	10	21	0.0300	0.0700	0.0000
28	10	22	0.0700	0.1500	0.0000
29	21	22	0.0100	0.0200	0.0000
30	15	23	0.1000	0.2000	0.0000
31	22	24	0.1200	0.1800	0.0000
32	23	24	0.1300	0.2700	0.0000
33	24	25	0.1900	0.3300	0.0000
34	25	26	0.2500	0.3800	0.0000
35	25	27	0.1100	0.2100	0.0000
36	28	27	0.0000	0.4000	0.0000
37	27	29	0.2200	0.4200	0.0000
38	27	30	0.3200	0.6000	0.0000
39	29	30	0.2400	0.4500	0.0000
40	8	28	0.0600	0.2000	0.0200
41	6	28	0.0200	0.0600	0.0100

Question. Solve using Newton–Raphson load flow method. Enforce PV-bus reactive-power limits with PV-to-PQ conversion if required. Report the converged bus voltages, generator powers, branch power flows, total real-power loss, iteration count, and final mismatch.